Varun Sahni

510-944-7378 | sahni8@purdue.edu | vsahni.me | github.com/notskamr | blog.vsahni.me

Education

Purdue University — West Lafayette, IN
B.S. Computer Engineering | GPA: 3.68 — *Aug. 2025 – May 2028*
- Relevant Coursework: Data Structures & Algorithms, Advanced C Programming, Digital System Design

University of California, Berkeley (Summer Sessions) — Berkeley, CA
Coursework in Computer Architecture & Data Science — *Summer 2023*
- Computer architecture: assembly, C, parallelism, pipelining, warehouse-scale compute

Experience

Paasa (YC S24) — Delhi, IN
Software Engineer Intern — *May 2026 – Aug. 2026*
- Saved 200+ ops hours/week by designing and deploying Paasa's AWS-hosted model-portfolio platform end to end, automating order placement through an audited execution layer, with Monte Carlo analytics & rebalancing
- Reduced time to a fully approved brokerage account from ∼1 week to 1 day by redesigning KYC onboarding to feed IBKR's API, automating account creation; shipped behind PostHog flags, owning monitoring & incidents
- Cut instrument ingestion from a 6-hour cron to minutes by rebuilding it on IBKR FTP feeds & OpenFIGI IDs (∼90K instruments), ending manual reconciliation; cut search latency from 1–2s to 100–200ms with Redis Search
- Designed and led modularization of cash systems (reconciliation & ledger) and a ∼10K-line monolithic trading service across ∼10 authored design docs as net AUM grew $15M to $80M on a 5-engineer team
- Lowered order latency from 2–3s over REST to 20–100ms in pre-production testing by building Paasa's first FIX trading engine in Kotlin — a QuickFIX core behind a typed gRPC/Protobuf API

Composable Silicon Lab — Purdue University — West Lafayette, IN
Undergraduate Researcher — *Sep. 2026 – Present*
- Working on RISC-V vector cores and ML accelerators under Prof. Vikram Jain, with a focus on RTL design

CortX — Remote / India
Co-Founder & Founding Engineer — *2022 – 2024*
- One of 5 founders; led a medical management platform (React, Express.js) to 1st at the ShriTeq Hackathon

Self-Directed Computer Vision Research — Remote
Researcher — *Jun. 2023 – Jul. 2024*
- Built a lightweight CNN for GI disorder detection (97.82% accuracy); first-author paper at IEEE ICSIGSYS 2026

Selected Projects

bun-sqlite-cache — Open-source caching library for Bun + SQLite (GitHub)
- In-memory TTL/LRU cache for Bun & SQLite benchmarked at ∼90K reads/s (∼11μs/read); 1K+ downloads

Flux — Live data-streaming proxy API (flux.vsahni.me | GitHub)
- Brings live SSE/WebSocket streaming to serverless apps; powers ∼200 actively streaming channels

Technical Skills

Languages: Java, Python, TypeScript, Go, Kotlin, Rust, C, SQL
Cloud & Infrastructure: AWS (incl. Bedrock), Docker, Kafka, gRPC/Protobuf, multithreading, OpenMP
Databases: PostgreSQL, MySQL, SQLite, MongoDB, Redis
Frameworks: React, Next.js, SvelteKit, Node.js, Bun, FastAPI, Express
Developer Tools: Git, GitHub Actions (CI/CD), PostHog, Claude Code, Cursor
Machine Learning: TensorFlow, Keras, NumPy, Pandas, Scikit-learn